

Education, Sex, and Income in the UCI Adult Dataset

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1 Dataset

We use the Adult dataset, extracted from the 1994 US Census database and retrieved on 29 September 2026 from the official UCI archive [1]. It contains **48,842 rows** and **15 columns**: 14 predictors and the binary target `income` (`>50K` or `<=50K`). The predictors mix continuous, discrete, and nominal attributes. `age` records respondent age in years. `education-num` records an ordinal education level, and `hours-per-week` records usual weekly work hours. `Workclass`, `occupation`, `sex`, and `race` describe employment and demographic groups. The raw files still contain `?` markers, exact duplicate records, and `capital-gain` values top-coded at USD 99,999. We use this dataset because its mixed feature types and documented data issues support both statistical inference and a reproducible preparation study.

Table 1. Selected attributes and their source meaning.

Attribute	Type	Unit or range	Real-world meaning	Missing
<code>age</code>	discrete	years, 17–90	Age of respondent	0%
<code>workclass</code>	nominal	8 known levels	Employer type	5.7%
<code>education-num</code>	discrete	ordinal, 1–16	Highest education level	0%
<code>occupation</code>	nominal	14 known levels	Job category	5.8%
<code>capital-gain</code>	continuous	USD, 0–99,999	Capital income; top-coded	0%
<code>hours-per-week</code>	continuous	hours, 1–99	Usual weekly work hours	0%
<code>sex</code>	nominal	Female or Male	Recorded sex group; sensitive attribute	0%
<code>fnlwgt</code>	continuous	positive integer	Census sampling weight	0%

The target has 37,155 records at or below USD 50,000 (76.1%) and 11,687 above USD 50,000 (23.9%). The full data dictionary appears in Appendix A.

2 Data preparation

The table records proposed steps. The encoding and predictor choices have not been applied to the full dataset or approved in the processing pipeline.

Table 2. Proposed preparation decisions, alternatives, and quantified reasons.

Issue	Proposed step	Alternative	Reason
Missing values (?)	Convert to missing, then retain an explicit unknown category	Drop rows or impute the mode	The source has 6,465 marker cells: 2,799 in work-class, 2,809 in occupation, and 857 in native-country. An unknown category preserves records and keeps missingness visible.
Duplicate rows	Retain exact duplicates	Drop repeated rows	Exploration contains 38 exact duplicates among 39,073 records. No record identifier shows they are errors, so dropping them would alter observed frequencies.
Capital-gain top-code	Keep and flag USD 99,999	Delete or winsorize capped values	There are 194 capped observations in exploration. The cap is a source limit, not evidence of invalid records.
Redundant education fields	Keep ordinal <code>education-num</code> ; omit <code>education</code> from the candidate model matrix	Include both fields	The 16 education labels map one-to-one to numeric levels in exploration, with 0 mapping conflicts. Including both repeats the same information.
<code>fnlwgt</code>	Exclude it as a personal predictor	Treat it as an individual attribute	<code>fnlwgt</code> is a census sampling weight, not a personal characteristic. Any population-weighted inference needs a separate review of the weighting design.
Encoding	Candidate: one-hot encode 7 nominal fields and retain 5 numeric or ordinal fields; use an explicit missing category	Integer-code nominal values	The candidate has 91 columns and 4,444,622 cells over 48,842 rows, above the 3,000,000 floor. Integer codes would impose an order on nominal categories. This encoding has not been applied.

3 Effect of preparation

Table 3. Projected effect of the proposed preparation. Duplicate counts refer to exploration only.

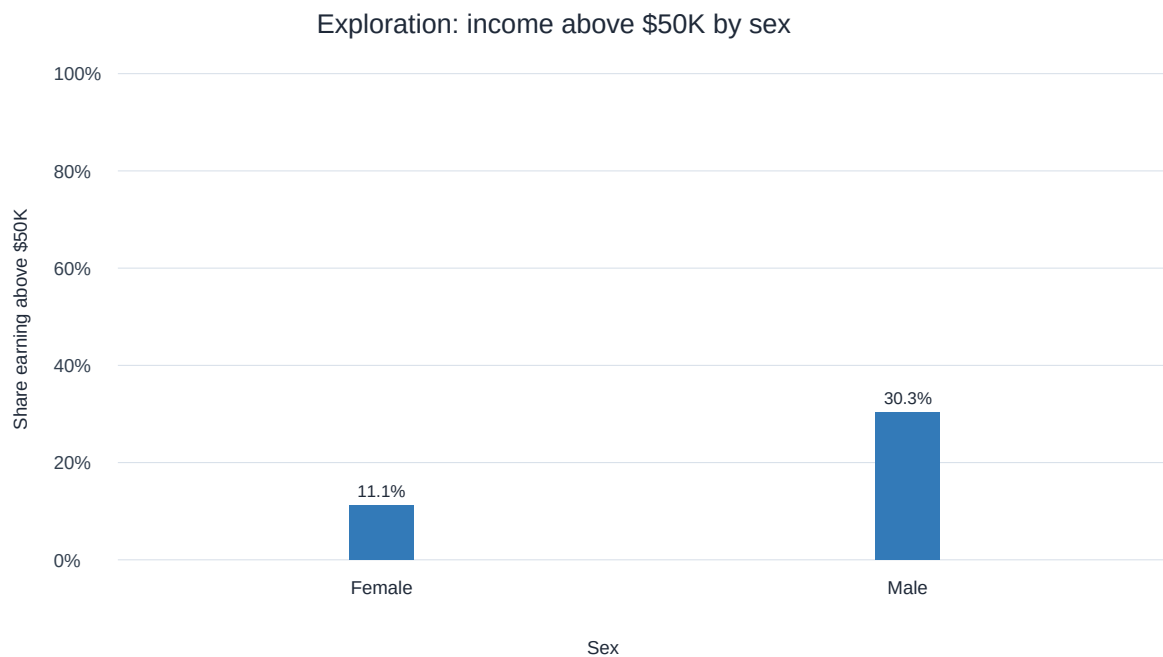
Metric	Before	Proposed after	Change
Rows	48,842	48,842	0
Predictor or encoded feature columns	14	91	+77
Missing-marker cells	6,465	0	−6,465
Exact duplicates in exploration	38	38	0
Share with income above USD 50,000	23.9%	23.9%	0.0 points
Samples × features	683,788	4,444,622	≥ 3,000,000: yes

The after column projects the proposed design. It is not a measured full-dataset processed artifact. The duplicate count covers the exploration portion because confirmation records remain closed. The proposal removes no rows. It omits two raw predictors from the candidate model matrix: the redundant `education` field and the sampling-weight field `fnlwgt`. It gains explicit missing categories and one column per nominal category. Source top-coding means the true capital-gain values above USD 99,999 cannot be recovered.

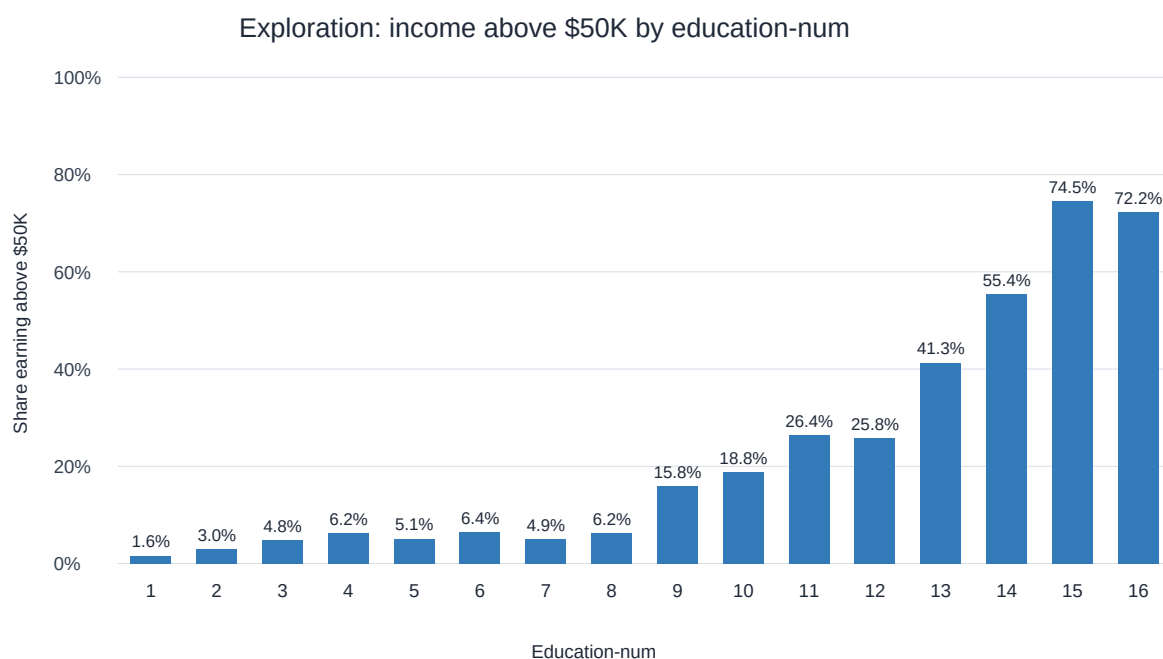
4 Exploration and confirmation split

We split before exploration and stratified on `income`. The exploration partition contains **39,073 rows (80.0%)** and the confirmation partition contains **9,769 rows (20.0%)**. The seed is **558**, defined once in `src/config.py`. All figures use exploration rows. Confirmation remains closed until the hypothesis and analysis choices are fixed.

5 Exploratory analysis



(a) The exploratory above-USD-50,000 rate is 19.1 percentage points higher for males. The rates are 30.3% (7,908/26,132) for males and 11.1% (1,441/12,941) for females.



(b) The above-USD-50,000 rate spans 72.9 percentage points across education levels. It ranges from 1.6% at level 1 (1/64) to 74.5% at level 15 (493/662).

Figure 1. Exploratory marginal rates motivate the proposed adjusted education-by-sex interaction analysis. These rates do not estimate adjusted effects.

5.1 Obvious questions

Anyone familiar with this dataset would ask:

- O1. Does income vary by education level?
- O2. Does the income distribution differ by sex?
- O3. Does income vary with hours worked per week?

5.2 Problem statement

P1. Primary. Does the association between `education-num` and income above USD 50,000 differ by sex? We will estimate it after adjusting for age and hours worked per week.

Why P1 is not on the obvious list. P1 asks whether the education association differs by sex after adjustment for age and work hours. The obvious questions ask only about marginal patterns. We will make one confirmatory claim to avoid unnecessary multiple testing.

5.3 Proposed inference

We will fit logistic regression with income above USD 50,000 as the binary outcome. Predictors will include `education-num`, `sex`, `age`, `hours-per-week`, and the `education-num-by-sex` interaction. The null hypothesis is that the interaction coefficient equals zero. The alternative is that it differs from zero. We will compare nested models with a one-degree-of-freedom likelihood ratio test and report the interaction estimate with a confidence interval.

This test assumes independent records, convergence, adequate cell counts, and linear effects of continuous terms on the log-odds. We will check convergence, sparse levels, and logit linearity. We will compare the result with a model that treats `education-num` as categorical and tests a 15-degree-of-freedom interaction. That alternative relaxes the linear education trend but uses more parameters. It is an assumption check, not a second confirmatory claim. No multiple-test correction is needed for one primary test; if later analyses add confirmatory tests, we will use Holm correction.

Sex, race, and native-country are sensitive attributes. We will report group counts and uncertainty and examine group differences in missingness. The observational data support association claims only. A causal claim about education would require a defensible causal design, temporal ordering, and a strategy for confounding. Group rate differences alone do not establish unfair treatment or measure fairness completely.

6 Reproducibility

The environment uses `uv`, `pyproject.toml`, and `uv.lock`. The split seed is **558**, and the raw archive remains unchanged. Run `uv run python -m src.data.prepare` to recreate the split and `uv run python -m src.analysis.explore` to regenerate the two proposal figures. The archive SHA-256 is `7537312dd56c2b9803580805ce99e68183a30ee4685aa5329d6df0fbb3cc21bb`. Project repository: <https://github.com/durgesh-k-sharma/cse558-adult-project>.

References

- [1] B. Becker and R. Kohavi. *Adult*. UCI Machine Learning Repository, 1996. <https://doi.org/10.24432/C5XW20>.

A Data dictionary

The missingness column reports source ? markers before cleaning.

Attribute	Type	Unit or range	Meaning or values	Missing
age	discrete	years, 17–90	Respondent age	0
workclass	nominal	8 categories	Employment class	2,799 (5.7%)
fnlwgt	continuous	positive integer	Census sampling weight	0
education	nominal	16 categories	Highest education category	0
education-num	discrete	1–16	Ordinal education level	0
marital-status	nominal	7 categories	Marital status	0
occupation	nominal	14 categories	Occupation class	2,809 (5.8%)
relationship	nominal	6 categories	Household relationship	0
race	nominal	5 categories	Recorded race; sensitive attribute	0
sex	nominal	2 categories	Female or Male; sensitive attribute	0
capital-gain	continuous	USD, 0–99,999	Capital gain; upper value is top-coded	0
capital-loss	continuous	USD, 0–4,356	Capital loss	0
hours-per-week	continuous	hours, 1–99	Usual weekly work hours	0
native-country	nominal	41 categories	Country of origin; sensitive attribute	857 (1.8%)
income	nominal	≤USD 50,000 or >USD 50,000	Income bracket; target	0